Eos Family Custom Encoder Maps Quick-Start Guide

Encoder Maps are a way to choose how parameters lay out on your encoders. You can use a Default Map for simple rearranging of parameters or go more advanced by creating specific maps for fixture types and letting Eos compile maps based on your channel selection.

Custom Encoder Maps are a mode. They are enabled by clicking/touching **{Custom}** in the encoder display. Once Custom is enabled, the current channel selection will determine which encoder map to use. To return back to the standard Eos mapping, simply toggle **{Custom}** off.

Before getting started, load any of the "Mind The Gap" Eos Family show files from the ETC Education Center materials.

Create an Encoder Map

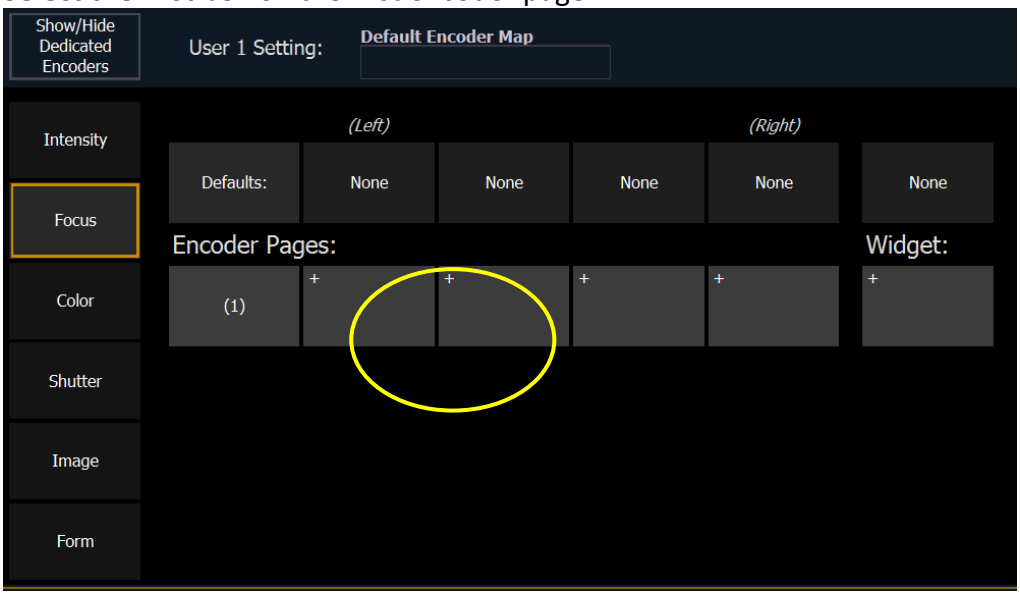
Add a new tab and select Encoder Maps (Tab 40)

[1] [Enter] will create a new map.

The boxes down the left side of the screen correspond with the encoder paging buttons on the console. Each category can have up to 25 pages of encoders. The 4 boxes arranged horizontally represent the 4 encoders within each page of the selected category.

Select the Focus category on the left side of the screen.

Select the first box on the first encoder page.



This will bring up the parameter selection pop-up window. Just like when using the fixture profile editor, we can choose from common parameters, look at all available parameters in any category, or use the search box.

Expand the Common list and then double tap or click on Pan. This will assign Pan to the first encoder. As soon as a page is populated with at least one parameter, a new row (page) of encoders will appear for use.

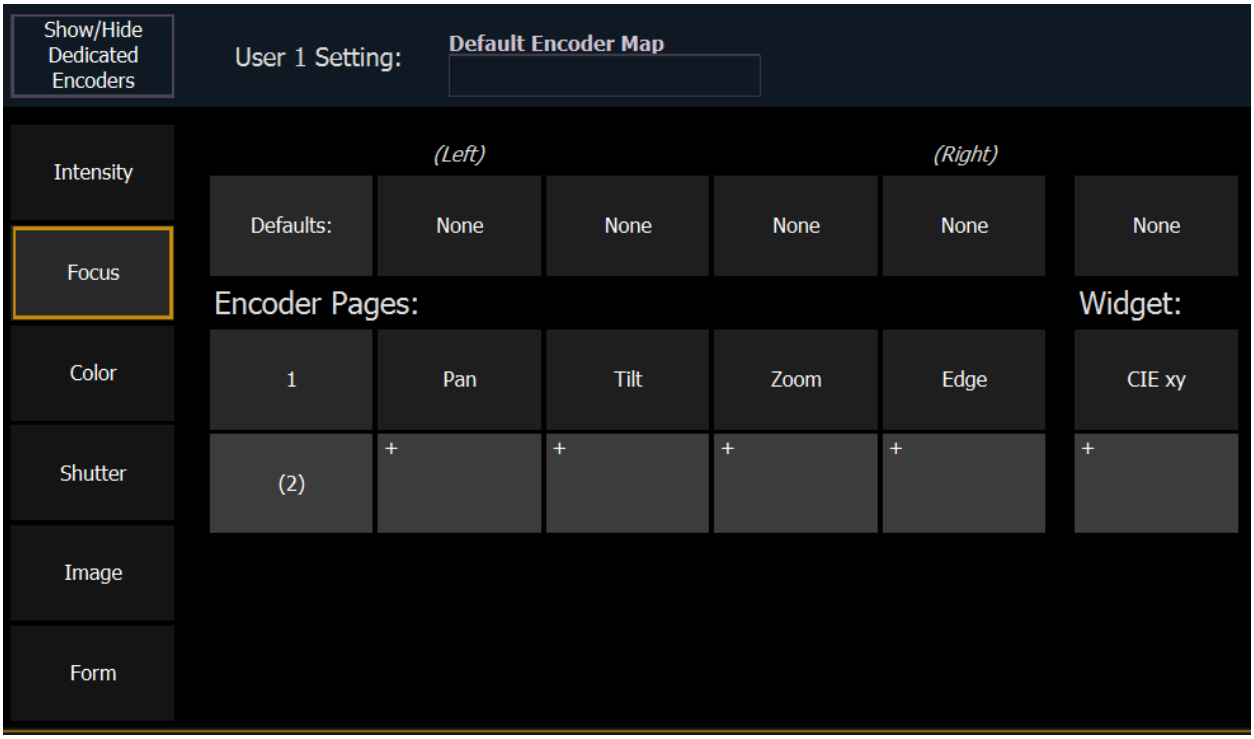
Now, **select the second box on the first row and select Tilt** from the Common menu.

Select the third box and this time, use the search box. Type in "Zoom" and double tap/click it to assign it to the third encoder.

Select the fourth box and use the search box to find "Edge" and assign it to the fourth encoder.

You can also choose to assign a Widget to each encoder page. **Select the Widget box (currently populated with "None") and under Color Pickers, choose "CIE xy"**

It should look like this when you are done:



Switch to the {Color} category. Click on the first box and find "Parameter Sets." Choose "RGB Group." Click on the first box of the second row (second page) and assign the "CMY Group" from Parameters Sets. On the 4th box for the first page assign "CTO." Also assign "CTO" to the 4th box on the 2nd page. It should look like this:



The console uses a predefined hierarchy to determine which encoder map to display at any given time. The standard map is what Eos uses if no custom maps exist or if custom maps are disabled. The next level up in the hierarchy is the Default Custom Map.

The Default Custom Map is defined at the top of the Encoder Maps Tab, or in the User > Manual Control section of Setup. This setting is specific to the current user.

Select the Default Encoder Map box at the top of the screen and type [1] [Enter] to assign this map as the default for User 1.

[Live], then type **[101] [Enter].**

Open the **[Encoder Display]**.

Push **[Focus]** to display the Focus parameters. Note that Pan and Tilt are assigned to the first 2 encoders, and the 3rd and 4th encoders are empty.

Enable Custom Encoders by pressing **{Custom}**

Now see that Zoom, Edge and the color picker are available and that the active map (Map 1) is displayed inside the Custom button:



Switch to the **[Color]** page of the encoders.

[51] [Enter] to select a ColorSource Spot. Note that the encoders are not populated now. This is because we are on the 2nd page of the color category (which was assigned at RGB, CTO) and this fixture does not have any of those parameters. Press **{Color}** to switch to the 3rd page of the color category. Note that Lime, Hue, and Saturation are now displayed.

When the default map is being used, Eos will automatically add all parameters that were not included in the map at the end of each category's pages. This way, all parameters are easily accessible when in the default map. Since we didn't assign those parameters in our default map, they have been automatically added to the last page of the category.

To deactivate the custom map, click/touch the **{Custom}** button. Eos will return to standard mapping.

Best Practices

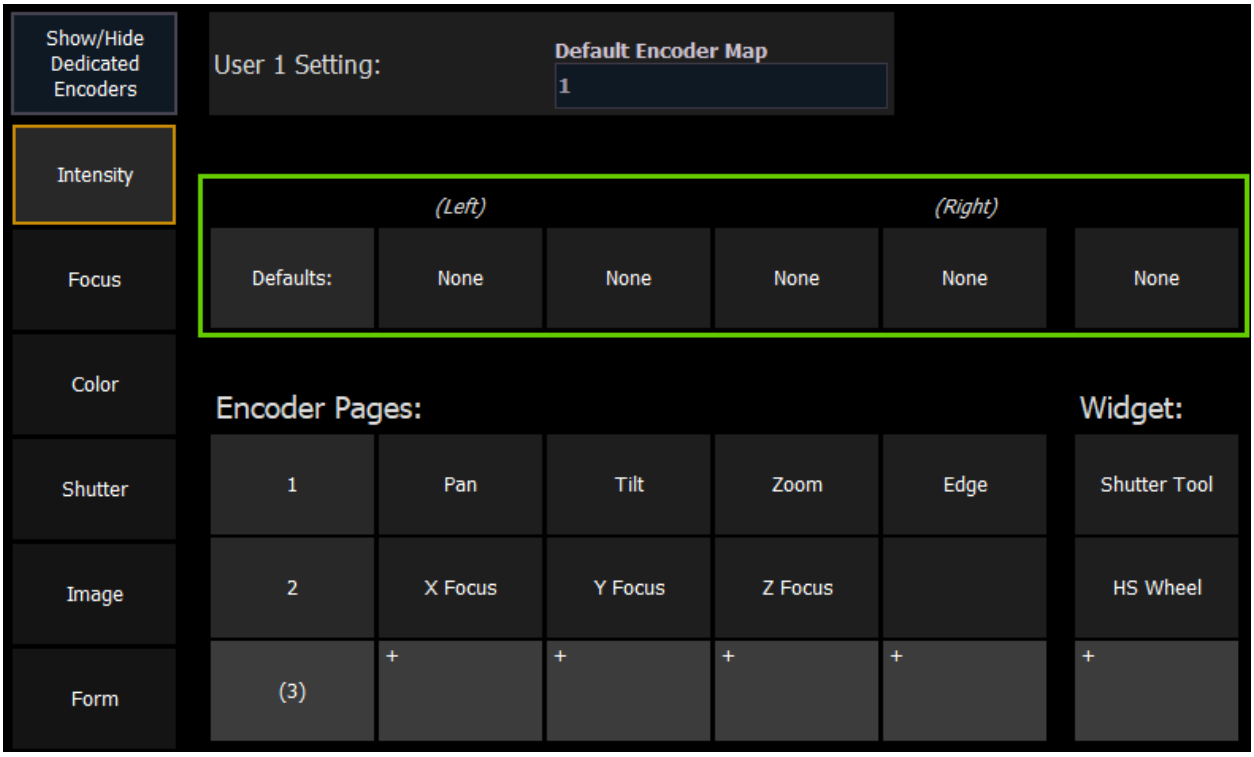
The Default Encoder Map setting can be changed with a macro. This can help change maps without relying on channel selection.

Default Encoder Map is a User setting, so changing users can also yield different default maps.

Default Parameters

Default parameters are used to quickly set a single parameter to always populate on the same encoder on each page within a category.

Go back to the Encoder Maps tab and select Map 1.



Select the Color category on the left, then click "CTO" on the 2nd row and choose "Clear."

Now, on the row of Default parameters at the top, click the 4th box and search for CTO and assign it. Note that the parameter name is displayed in blue, and the 4th box on the 2nd page has populated with "CTO" in a magenta color. This color scheme matches the by-type palette scheme in that the "leader" parameter is blue and the "follower" parameters are magenta.

Assign "Blue" to the 1st parameter on the 3rd page. Once a page has at least one parameter assigned to it, the default parameters will also populate.

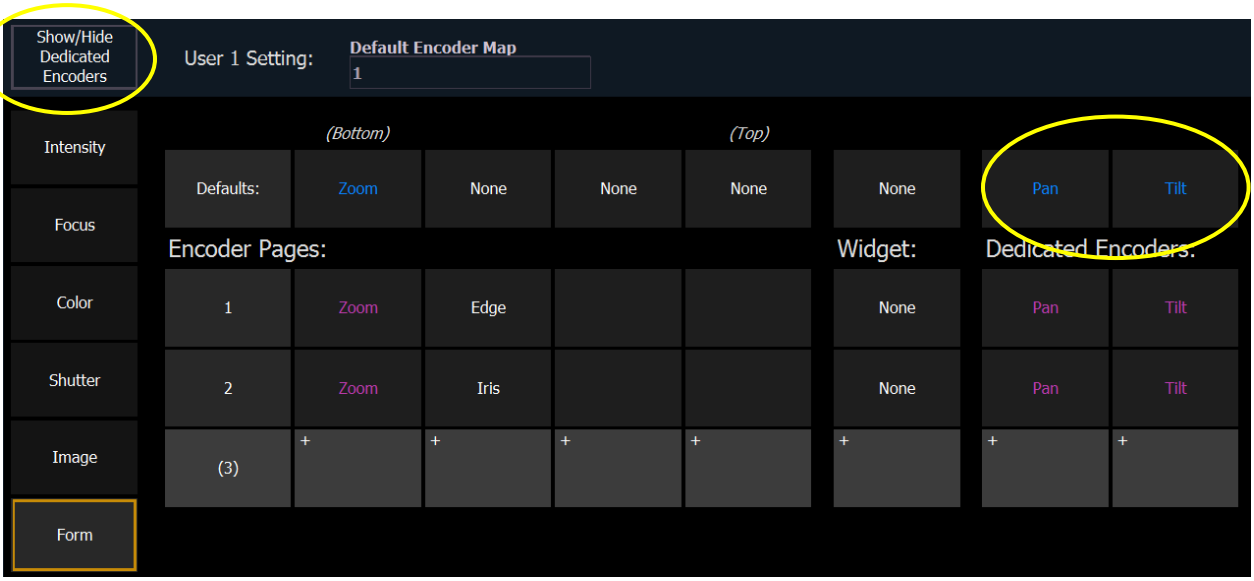


Click on the white "CTO" on the first page choose "Clear." Like with by-type palettes, removing this discrete data will cause the parameter to revert back to the default parameter.

To override the default parameter, you can choose any parameter and select a new one. This parameter will display in white – like discrete data in a by-type palette. **Select "CTO" on the third page and replace it with "Shutter Strobe"**

We can also delete an entire row of encoders. **Click on the (3) indicating page 3 and select "Delete"** to remove the whole row.

For consoles with a dedicated encoder touchscreen and 6 encoders, we can also map the extra pair of dedicated encoders. Press **{Show/Hide Dedicated Encoders}** to toggle visibility of these encoders:

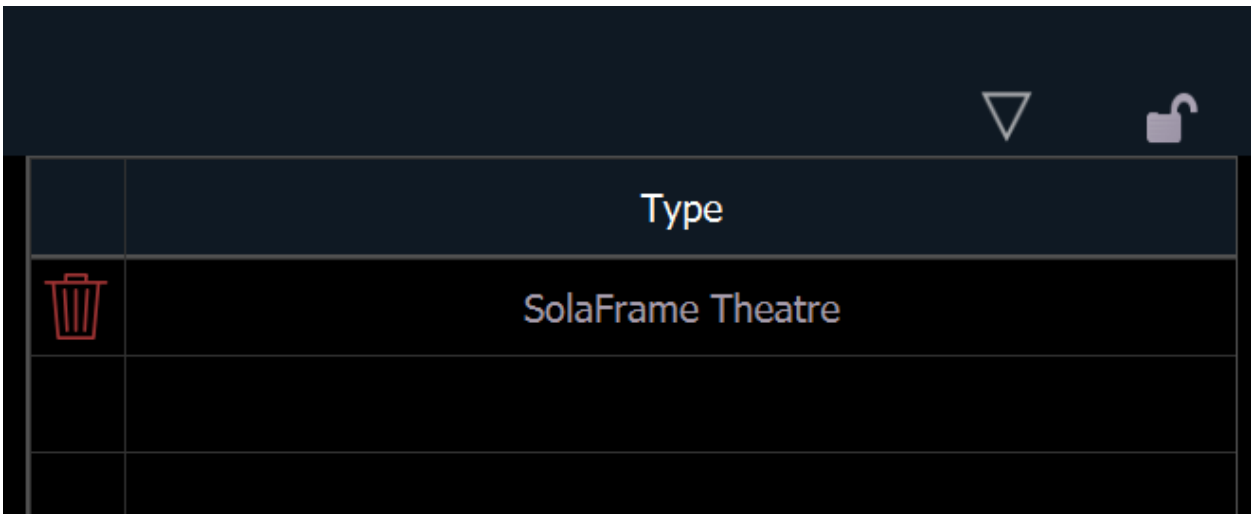


The dedicated encoders default to Pan/Tilt on every category but can be changed the same way as all other parameters.

Create from Fixture & Map to Fixture

The next level in the hierarchy of which encoder map Eos will display are maps assigned to specific fixtures. Fixtures are assigned to maps in the Encoder Maps tab.

Type **[2] {Create From Fixture} and select SolaFrame Theatre**. It is possible to search the fixture database for any fixture type, but by default, Eos displays fixtures that are already patched into your show file. This will create a full encoder map that matches the standard Eos mapping. You can now edit this map rather than having to build the whole thing from scratch. Eos automatically associates the fixture type from which the map was created to this map. It is possible to map additional fixture types to any map by using the {Map to Fixture} softkey. To unassign a fixture type from an encoder map, use the {Map to Fixture} softkey, then click the red trashcan next to the fixture type you wish to unassign:



Press [Label] and type SolaFrame Theatre

Go to the Intensity category on the left. Select the 2nd, 3rd, and 4th parameters and Clear each one. It can be useful to clean up the encoders by removing parameters that are not often needed.

Now, go to the Form category. Click on the (2) of the 2nd row and click "Delete." Next, select the 4th parameter which is currently "Shutter Strobe," select Change Parameter then search for and select Diffusion. Assign "Shutter Strobe" to the first encoder on the 2nd page. This is an example of rearranging parameters to match a personal preference.



[3] [Enter] [Label] ColorSource Spot [Enter] {Map to Fixture} ColorSource Spot Direct

Select the Color Category. Click the (1) for the first page and select Clear. This is different from Delete in that it removes the categories, but does not delete the page. **Now click on the first parameter box and go to Parameter Sets. Choose the RGB Group. Click the 4th box and search for "Lime" and assign it. Now, Delete the 2nd row of parameters.**

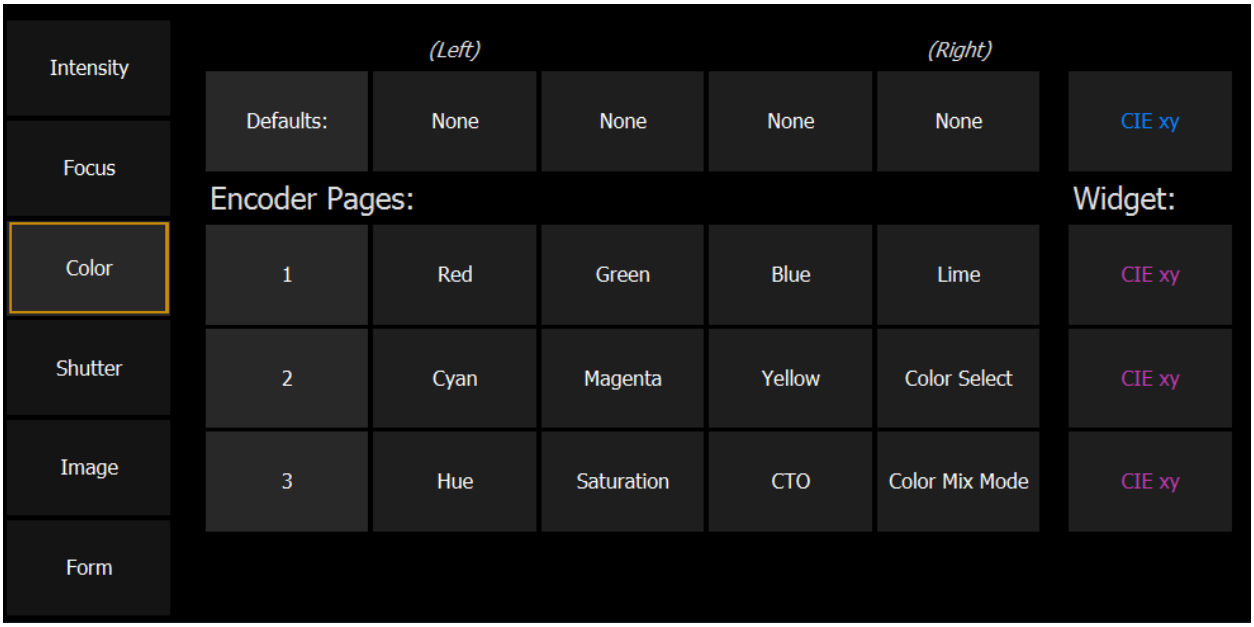
Select the (1) on the first page and choose Copy. Now go to the Focus parameters and click the (1) for the first page and choose Paste. Do the same thing on the Shutter and Image categories. We have now assigned the color parameters to these formerly unused categories so they are available no matter which category is currently assigned to our encoders.

[2] [Copy To] [4] [Enter] [Label] SF Theatre & CS Spots

{Map to Fixture} ColorSource Spot DB

The SolaFrame Theatre was already assigned since we started from the copy of Map 2. Here we have assigned a 2nd fixture type to use with this map. You can associate as many fixtures to a map as you like, but keep in mind that Eos will only use a map when the fixture selection contains all fixture types that have been associated with that map.

Go to the Color category, click the (1) and choose Insert Before. Now, go select map 3 and choose (1) then Copy the row of parameters. Come back to Map 4 and Paste those parameters onto the empty row.



The Encoder Mapping Hierarchy

With Custom Mode enabled, the console will examine your channel selection and then go through a process to determine which map to use. First, the console will look for an exact match – meaning, are all the fixture types in your selection assigned to a single map. This is the highest level in the hierarchy whether you have one or multiple fixture types selected.

[Live]

[51] [Enter] This is a single fixture type which is assigned to Map 3.

Open Encoder Displays and look at the Color category. You will see Red, Lime, Green, Blue on page 1 and Hue, Saturation on page 2. This is the standard Eos mapping.

{Custom} to enable custom maps. See that the map label appears within the Custom button – in this case it is map 3 since our channel selection only contains the fixture type that is assigned to map 3. Now the encoders are populated with Red, Green, Blue, Lime and there is only a single page of encoders in the color category. Note that with the default map, unassigned parameters were added to the end of each category – this is not the case with maps assigned to specific fixtures.

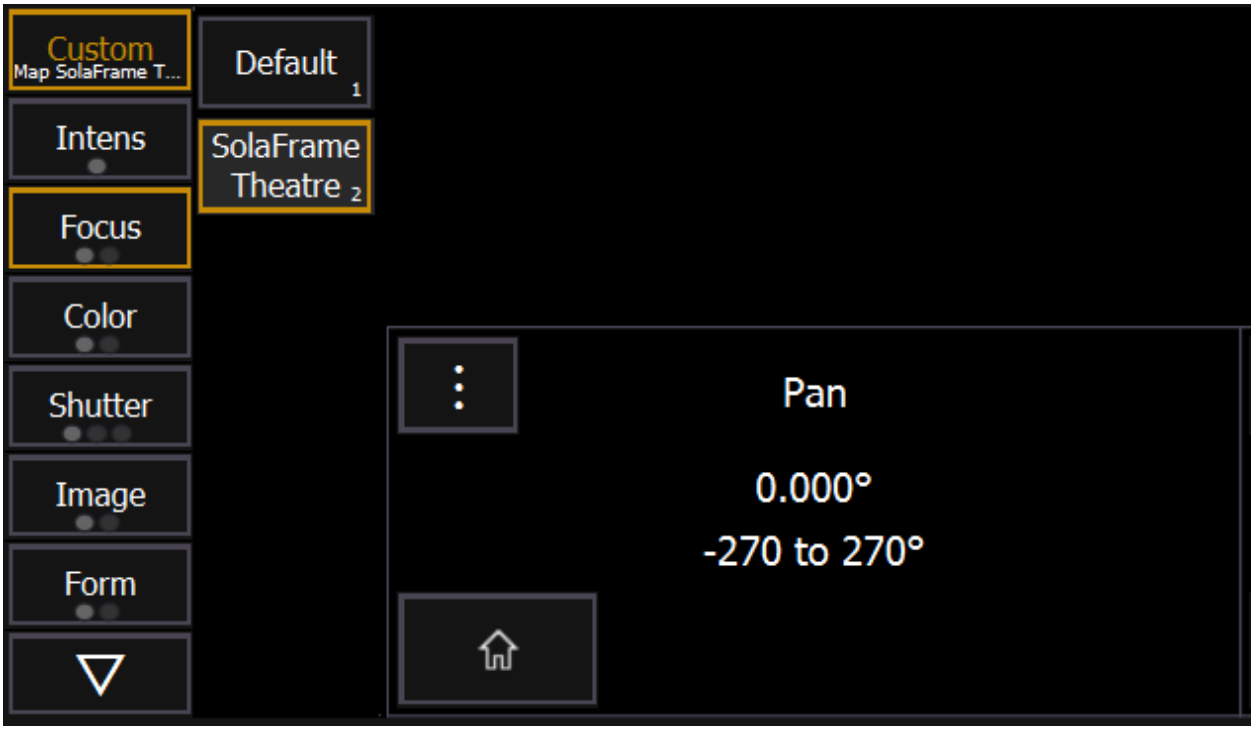
Switch to Focus, then Shutter, then Image categories. Each one has the same parameters because that is how we laid out Map 3.

[51] [+] [101] [Enter] to select a ColorSource Spot and a SolaFrame Theatre. Custom Maps are still enabled, but now the console is displaying Map 4 based on our channel selection of those 2 fixture types. This is also an exact match, since Map 4 has these two fixtures assigned to it. **Look at the Color category of encoders**. Note that the first page is Red, Green, Blue, Lime and the rest of the pages are the SolaFrame Theatre parameters, just as we built it.

[101] [+] [121] [Enter] This is a partial match – the next level down in the hierarchy. This selection contains a fixture type that is assigned to a map (SolaFrame Theatre) and a fixture type that isn't assigned to a map (SolaWash 2000). In the encoder display, you'll see map filter buttons appear on the left – one for each map being used. In this case, because we have a SolaFrame Theatre within our selection, Map 2 is an option, and because we have a fixture that isn't assigned to any map, Map 1 the default, is the other option. Eos will "stack" these maps in each category, in the order of Map 1 page 1; Map 1 page 2; Map 2 page 1, Map 2 page 2; etc. If you touch/click on a map filter, it will hide all other pages and only show you that map. To return to all maps, touch the selected map filter to deselect it.

Eos will "prefer" a map that is associated with all fixtures (as was the case when we selected 51 & 101) before it stacks maps associated with individual fixture types.

Eos uses some rules to try to present the fewest maps possible. If a map has an exact match of fixture types, it will present that map. If no such map exists, Eos will look for a map with the *most* fixtures matching, and present that map. With any leftover fixtures, Eos will look for another match or match with the most fixtures. This continues until all channels are matched to maps, and then Eos "stacks" the maps and presents map filters. If there are channels that aren't associated with any maps, the default is added to the stack too.



[121] [Enter] to select a SolaWash 2000. This fixture type is not assigned to any encoder map, so the console will use the default map (Map 1) – this is the lowest level of the custom encoder hierarchy. Look at the Color category. The first page only contains CTO because the map was built with Red, Green, Blue as the first three parameters – they are not shown because this fixture doesn't have those parameters. Look at the Shutter, Image and Form categories – even though we left these blank in the default map, there are parameters populated here; this is a special feature of the default map – the console will place unassigned parameters at the end of a category The default map always gives you access to all parameters even if they are not specifically assigned in the map.

{Custom} We have now disabled custom maps and are back to standard Eos encoder mapping.

Note: in the event that multiple maps meet the same criteria for use, the console will use the lowest numbered map.

Filter Maps by User

Default Encoder Maps are assigned per user, and individual maps can each be assigned to a specific user or users.

By default, new maps are available to all users, and Eos will match a channel selection with them regardless of which user is making the selection. By applying user filters to a map, only the user(s) associated with the map will be eligible to match channel selections to them.

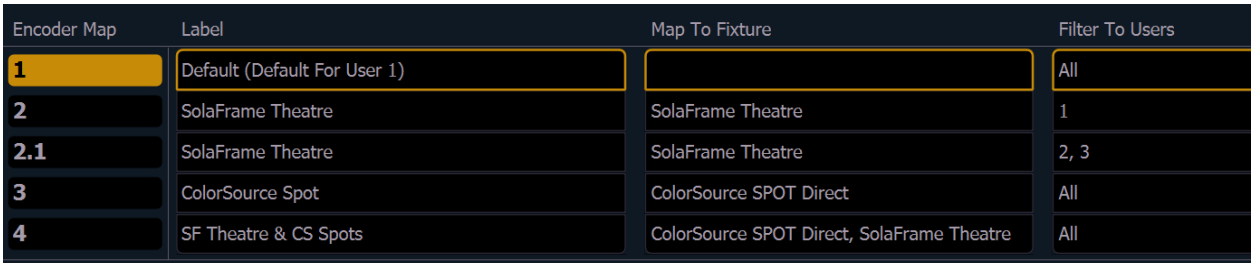
This makes it easy to delineate swaths of maps between users, so that each person can customize how they interact with each fixture type. Once user filters are applied, all channel selection matching rules are utilized to choose the map(s) to present.

In the encoder maps tab

[2] [Copy To] [2.1] [Enter]

With Map 2 selected, {Map to Users} [1] [Enter]

Select Map 2.1, {Map to Users} [2] [Thru] [3] [Enter]



Encoder Map	Label	Map To Fixture	Filter To Users
1	Default (Default For User 1)		All
2	SolaFrame Theatre	SolaFrame Theatre	1
2.1	SolaFrame Theatre	SolaFrame Theatre	2, 3
3	ColorSource Spot	ColorSource SPOT Direct	All
4	SF Theatre & CS Spots	ColorSource SPOT Direct, SolaFrame Theatre	All

Given these maps and User Filters, here are some examples of which custom maps will be used:

User	Channel Selection	Map
1	SolaFrame Theatre	2
1	ColorSource Spot	3
2	SolaFrame Theatre	2.1
2	ColorSource Spot	3
3	SolaFrame Theatre	2.1
4	SolaFrame Theatre	1
4	ColorSource Spot	3

Note that when User 4 selects a SolaFrame Theatre, the default map (Map 1) is used since User 4 is not assigned to any of the SolaFrame Theatre maps. To remove a User Filter, select a map, press {Filter to Users} and press enter. This will change the User Filter to “All.”